

LE-PAVD: Learning-Enhanced Physics-Aware Vehicle Dynamics for High-Speed Autonomous Navigation

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Abstract—Accurate modeling of nonlinear vehicle dynamics is essential for high-speed autonomous racing, where controllers operate at the handling limits. Model-based methods are interpretable but rely on simplifying assumptions, while purely learned models capture nonlinearities yet often lack physical consistency and generalization. We propose LE-PAVD (Learning-Enhanced Physics-Aware Vehicle Dynamics), a hybrid model that integrates physics priors with learned components. Our architecture adds four components: load-sensitive Pacejka tire forces, longitudinal load transfer, lateral tire-force effects, and rate-limited actuator inputs. Trained end-to-end on simulation and real-world telemetry, LE-PAVD enforces physical consistency while improving state prediction accuracy. On an unseen track, LE-PAVD reduces average displacement error (ADE) by 16.1%, final displacement error (FDE) by 20.6%, and lowers yaw-rate root mean squared error (RMSE) by 91.3% versus a deep dynamics baseline, while using 21.6% fewer FLOPs and achieving $\sim 1.50\times$ faster inference. In closed-loop simulations, LE-PAVD consistently outperforms the baseline by achieving faster lap times by 17.4% on a training track and 9.5% on a test track, without any track boundary violations. Overall, LE-PAVD offers a compact, physics-grounded dynamics backbone that improves predictive fidelity and closed-loop performance while reducing inference cost.

I. INTRODUCTION

Autonomous racing is a high-speed testbed for advances in control, planning, and vehicle dynamics. [1], [2]. Unlike conventional autonomous driving, racing vehicles operate under extreme conditions, including sharp cornering, rapid acceleration, and high-G braking, where maneuvers press against limits of tire grip, actuator response, and vehicle stability [3]. Accurate, robust, and efficient dynamics modeling is essential for predictive control, trajectory optimization, and real-time decision-making [4]. Beyond racing itself, insights gained here directly transfer to broader autonomous driving applications where reliability and safety margins are critical.

Traditional vehicle models (single-track and double-track) are interpretable but rely on simplifying assumptions such as constant stiffness, quasi-static loads, or instantaneous actuation. These assumptions break down in high-slip and transient

regimes, and identifying parameters requires costly instrumentation [5]. By contrast, deep learning models capture nonlinearities directly from telemetry [6], [7], but often lack interpretability, generalization, and safety guarantees. Such models can yield physically infeasible predictions that destabilize closed-loop control [8]. This persistent trade-off motivates hybrid models that combine physical consistency with learning-based flexibility.

In this work, we propose the Learning-Enhanced Physics-Aware Vehicle Dynamics (LE-PAVD) model. LE-PAVD integrates physics priors with neural components by explicitly modeling (i) load-sensitive Pacejka tire forces, (ii) longitudinal load transfer, (iii) lateral tire-force variation, and (iv) rate-limited actuator inputs.

Trained end-to-end on telemetry from simulation and real-world, LE-PAVD enforces physical consistency while delivering accurate, lightweight predictions. We evaluate LE-PAVD on simulated and full-scale racetrack data in open-loop prediction. On real-world telemetry, LE-PAVD reduces trajectory prediction error by about 20% while using fewer computational resources.

We also show that adding physical structure yields larger closed-loop gains with a shallower network, even under identical controllers. Together, these results support LE-PAVD as a practical, physics-aware foundation for faster, safer, and more resource-efficient autonomous racing.

II. RELATED WORK

Estimating vehicle dynamics for autonomous racing has been studied extensively across both scaled and full-scale platforms. Existing approaches fall into three families: (i) classical physics-based models, (ii) data-driven neural models, and (iii) hybrid or physics-informed learning architectures. We emphasize two primary baselines-DDM [9] and FTHD [10]-as closest to our setting; contrasts below clarify where LE-PAVD extends them.

A. Physics-Based Vehicle Dynamics Models

Classical dynamics models use Newtonian mechanics to describe system behavior. Kinematic bicycle models are widely adopted for planning due to simplicity but neglect inertial and frictional effects [11]. Dynamic single- and two-track models incorporate lateral/longitudinal forces, inertial coupling, and tire

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slip dynamics [12]. Multi-DoF models capture suspension and load transfer but require extensive parameter identification and computation [13]. High-fidelity Pacejka-based formulations improve tire modeling but demand calibration and instrumentation [5]. Most physics-based models assume static parameters and omit actuator delays or suspension transients, limiting accuracy under near-limit racing transients [14]. These models provide interpretability but struggle with near-limit transients (load sensitivity, damping, delay) that LE-PAVD embeds directly into training and inference.

B. Data-Driven and Deep Learning Models

To bypass parameter identification, supervised learning approximates vehicle dynamics from telemetry [15]. Neural models capture nonlinear effects and often outperform classical models in-distribution [6], [16]; e.g., [16] pairs a differentiable UKF (unscented Kalman filter) with a learned model and online friction estimation for aggressive maneuvers. However, black-box models extrapolate poorly and provide limited physical insight. End-to-end policy learning approaches such as DeepRacing [17] achieve strong performance but do not explicitly learn vehicle dynamics models and are therefore not suitable as dynamics backbones for model-based control. Recent work explores Gaussian Process RL with collision avoidance [18] and digital-twin contextual RL [19], but interpretability remains limited. LE-PAVD retains learning capacity while constraining parameters and forces with physics priors to improve generalization and control-readiness.

C. Hybrid Learning Models

Hybrid frameworks embed physical priors within neural architectures to combine interpretability and expressiveness. The Deep Dynamics Model (DDM) [9] uses a Physics-Constrained Neural Network to estimate Pacejka coefficients and drivetrain resistances within a single-track model. While effective, it assumes fixed tire parameters and neglects load-dependent tire forces, nominal tire-force variation, suspension effects, and actuator delays. These omissions limit its ability to capture the nonlinear dynamics critical for aggressive maneuvers in autonomous racing. The Fine-Tuning Hybrid Dynamics (FTHD) model [10] augments a static physics model with a residual PINN and incorporates an EKF layer for denoising, enhancing

robustness under limited data. However, it adopts a sequential training approach rather than end-to-end optimization and treats the physics model as fixed during learning. Moreover, it omits key dynamic effects such as load transfer, suspension damping, and actuator latency, limiting its fidelity in high-slip and transient conditions. Residual learning strategies augment calibrated single-track models with Gaussian-process residuals inside MPC [20]. Broader approaches include PINNs for slip-induced dynamics [21] and adaptive Koopman operators with acceleration-based constraints and online adaptation [22]. LE-PAVD extends the hybrid line by *explicitly modeling* load sensitivity and longitudinal load transfer, while treating suspension effects as transient nominal tire-force variation and actuator inputs as rate-limited commands. Feasible parameter ranges are enforced through the physics guard layer during end-to-end training.

In summary, Physics-based models offer interpretability but oversimplify transient effects. Purely neural approaches capture nonlinearities but fail to generalize safely. Hybrid methods highlight the value of embedding priors but still omit load sensitivity, suspension dynamics, and actuator delays. These gaps motivate the need for a more expressive, physics-aware dynamics formulation, which we formalize next in Sec. III.

III. PROBLEM STATEMENT

We formalize the challenge of accurately modeling the dynamics of an autonomous racecar for real-time prediction and control. Realistic vehicle modeling is critical for downstream tasks such as model predictive control (MPC), trajectory tracking, and reinforcement learning-based decision making.

A. Physics-Aware Dynamic Single-Track Model

The continuous-time state evolution of the single-track model is governed by

$$\dot{v}_x = \frac{1}{m} (F_{rx} - F_{fy} \sin \delta + m v_y \omega), \quad (1)$$

$$\dot{v}_y = \frac{1}{m} (F_{fy} \cos \delta + F_{ry} - m v_x \omega), \quad (2)$$

$$\dot{\omega} = \frac{1}{I_z} (l_f F_{fy} \cos \delta - l_r F_{ry}), \quad (3)$$

where m denotes the vehicle mass, I_z is the yaw moment of inertia, and l_f , l_r are the distances from the center of gravity to the front and rear axles, respectively. The overall single-track formulation, including slip angles, normal loads, tire forces, and state vectors are illustrated in Fig. 1 and Table I details all the parameters described in Eq. (1)-(3). Building on the standard bicycle model, we propose an enhanced physics-aware single-track model that includes:

- **Nonlinear tire forces** using load-dependent Pacejka “Magic Formula” with dynamic normal force scaling (Sec. IV-B1).
- **Dynamic load transfer** effects in the longitudinal directions due to braking, acceleration, and lateral effects due to cornering (Sec. IV-B2).
- **Lateral tire forces** that capture variations in normal force that are otherwise neglected in simple planar models (Sec. IV-B3).

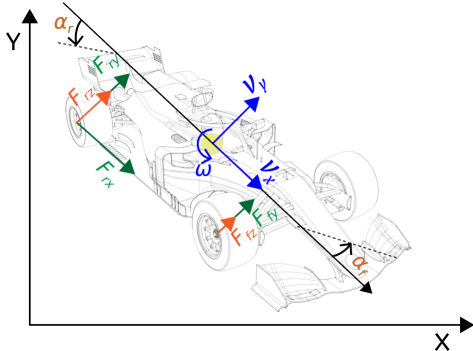


Fig. 1. Single-track vehicle model used in LE-PAVD. State information shown as $[v_x, v_y, \omega]^T$, along with slip angles α_f and α_r , normal loads F_{fz} and F_{rz} computed with longitudinal load transfer, and tire forces F_{fx} , F_{fy} , F_{rx} , F_{ry} . This baseline structure is extended in our proposed framework.

- **Rate-limited actuator inputs** (learned via the recurrent module) to reflect the finite response of steering and throttle in real systems (Sec. IV-B4).

TABLE I
STATE, CONTROL, AND DYNAMICS PARAMETERS USED IN THE LE-PAVD

State Variables		
Symbol	Description	Units
x, y	Global position in world frame	m
θ	Yaw angle (heading)	rad
v_x	Longitudinal velocity (body frame)	m s^{-1}
v_y	Lateral velocity (body frame)	m s^{-1}
ω	Yaw rate	rad s^{-1}
Control Inputs		
Symbol	Description	Units
T	Throttle input	%
δ	Steering angle input	rad
ΔT	Throttle change	%
$\Delta \delta$	Steering change	rad
Dynamics and Vehicle Parameters		
Symbol	Description	Units
l_f, l_r	Distance from CG to front/rear axle	m
h_{cg}	Height of center of gravity	m
m	Vehicle mass	kg
I_z	Yaw moment of inertia	kg m^2
F_{rx}	Longitudinal force on rear tires	N
F_{fy}, F_{ry}	Lateral force (front/rear)	N
F_{fz}, F_{rz}	Normal force on front/rear tires	N
F_{z0}	Nominal static normal force	N
C_{m1}, C_{m2}	Throttle gain coefficients	–
C_{r0}, C_{r2}	Rolling resistance coefficients	–
B_f, C_f, D_f, E_f	Pacejka tire parameters (front)	–
B_r, C_r, D_r, E_r	Pacejka tire parameters (rear)	–
S_{lf}, S_{hr}	Front/rear slip angle shift	rad
S_{vf}, S_{vr}	Front/rear lateral force shift	N

The state vector $\mathbf{X}_t = [v_x, v_y, \omega]^\top$ consists of longitudinal velocity, lateral velocity, yaw rate. The actuator commands are modeled as rate-limited changes of $\mathbf{U}_t = [T, \delta]^\top$. Given commanded increments $[\Delta T_t^{\text{cmd}}, \Delta \delta_t^{\text{cmd}}]^\top$, we obtain *applied* inputs $\tilde{\mathbf{U}} = [\tilde{T}_t, \tilde{\delta}_t]^\top$ via the learned temporal mapping in Sec. IV-B4, which are then used in the dynamics.

B. Motivation for Model Improvement

Despite progress in vehicle dynamics modeling, current approaches including the DPM [8], DDM [9] and FTHD [10] face fundamental limitations that hinder real-time control and long-horizon prediction tasks. DPM [8] relies on empirical coefficients that lack physical interpretability and adaptability. It does not inherently account for dynamic phenomena like load transfer or actuator delay-key factors in high-speed or aggressive driving. While prior models like DDM [9] and FTHD [10] introduced physics-guided learning, they omit key transient dynamics such as load-sensitive tire forces, suspension smoothing, and actuation delay-phenomena that dominate under limit-handling conditions. This motivates a more expressive model architecture, which we now formalize.

Our LE-PAVD model directly addresses these limitations through a tightly integrated architecture that blends interpretable physics with deep learning (See Fig. 2). Unlike previous models,

LE-PAVD performs end-to-end learning of physically meaningful parameters within an explicit dynamics model, allowing it to adaptively capture nonlinearities in high-performance driving. It embeds physics directly into the network structure, modeling load-sensitive tire forces, transient nominal tire-force variation via first-order dynamics, and rate-limited actuator inputs as native components of the system. The learned dynamics are integrated using a custom Euler scheme that incorporates time-varying actuator and suspension effects, enabling stable multi-step rollout under transient conditions.

Training minimizes one-step state prediction error subject to bounded physical parameter constraints, while model performance is evaluated using root-mean-square error (RMSE), average displacement error (ADE), and final displacement error (FDE). The model also demonstrates real-time inference capability, enabling seamless deployment in closed-loop systems, including model predictive control (MPC) and deep reinforcement learning (DRL) pipelines.

In this work, we limit our primary comparison to DDM [9], as FTHD [10] is neither evaluated for real-time deployment nor designed for control integration, and lacks support for critical dynamic effects modeled in LE-PAVD. LE-PAVD work focuses on delivering significant improvements over DDM by offering greater robustness, long-horizon consistency, and integration-readiness for control-critical autonomous driving tasks.

C. Physics-Informed Learning Objective

The model is trained by minimizing the prediction error between the estimated and observed future states $\mathbf{X}_{t+1}$, using only the current state and control input, while enforcing the learned dynamics through the physics-based differential equations [9]. The learning objective is defined as:

$$\hat{\Phi}_u = \arg \min_{\Phi} \frac{1}{N} \sum_{t=1}^N \left\| \mathbf{X}_{t+1}^{\text{pred}} - \mathbf{X}_{t+1}^{\text{obs}} \right\|^2 \quad \text{s.t. } \Phi_{\min} \leq \Phi \leq \Phi_{\max} \quad (4)$$

which includes learning the tire and drive train coefficients using a physics-informed neural network (PINN) that incorporates a *Physics Guard Layer* (See Fig. 2) to ensure all the predicted parameters stay within the physically meaningful bounds. This physics guard layer constrains the learned physical parameters via clamping within empirically defined bounds (See Table III), ensuring physical plausibility during both training and inference.

IV. IMPROVED VEHICLE DYNAMICS MODEL

A. Comparison of Model Components

To clearly illustrate the enhancements introduced in our physics-based vehicle model, Tables I and II summarize both variable-level and model-level improvements. Table II presents a side-by-side comparison between the original DDM [9] and our LE-PAVD across major components of vehicle dynamics modeling. Our modifications incorporate physically grounded formulations, including load-sensitive lateral forces, improved slip angle computation, transient nominal tire-force variation, and rate-limited actuator inputs.

In addition, Fig. 2 illustrates the overall LE-PAVD architecture. Sequential state and control inputs are first encoded by a

TABLE II
COMPARISON OF MODEL IMPROVEMENT WITH DDM AND OUR PROPOSED LE-PAVD

Component	Deep Dynamics Model (DDM)	Proposed - Improved Model (LE-PAVD)
Longitudinal Force (F_{rx})	$(C_{m1} - C_{m2}v_x)T - C_{r0} - C_{r2}v_x^2$	Same longitudinal force law as DDM, evaluated within a load-sensitive vehicle dynamics model with dynamic normal load variation (Eq. (8))
Axle Load Transfer (F_{fz}, F_{rz})	Not modeled	Modeled via load transfer effect (Eq. (9) & Eq. (10))
Lateral Tire Forces (F_{fy}, F_{ry})	Magic Formula evaluated at nominal normal load, neglecting dynamic load transfer effects	Lateral tire forces are computed using instantaneous normal loads F_z , enabling load-sensitive tire dynamics and capturing nominal load transfer effects (Eq. (11) & Eq. (12))
Actuation Delay Modeling	Not supported	Learned rate-limited behavior via recurrent temporal encoding; no hand-tuned saturation or lag Eq. (17))
Network Architecture	Heavy Neural Network	Introducing enhanced vehicle dynamics enables accurate modeling with a shallower network, reducing parameter count and computational overhead.

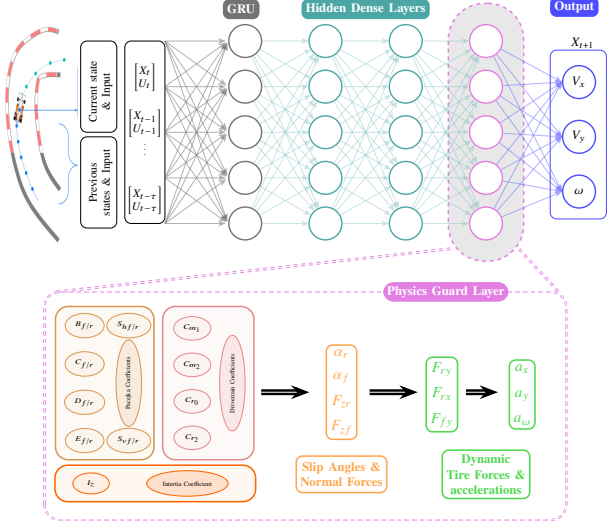


Fig. 2. LE-PAVD architecture. Sequential feedback and commanded control inputs are processed by a GRU and dense layers to predict next-step states $[v_x, v_y, \omega]^T$. A Physics Guard Layer enforces parameter bounds and computes interpretable dynamics through slip angles, normal loads, and tire/longitudinal forces, and accelerations while capturing rate-limited actuator behavior.

GRU network, followed by hidden dense layers that map temporal features to next-step predictions of (v_x, v_y, ω) . A physics guard layer constrains learned parameters within physically valid ranges and embeds domain knowledge by computing slip angles, normal loads, and tire/longitudinal forces, which are then applied to update the vehicle dynamics. This integration of neural sequence modeling with physics-based constraints ensures both predictive accuracy and interpretability. LE-PAVD architecture for the real-world dataset $\mathcal{R}_D$ adopts a recurrent, physics-aware neural architecture designed to encode temporal context for nonlinear vehicle dynamics with high temporal fidelity and physical interpretability. The network begins with a five-layer gated recurrent unit (GRU) comprising 144 hidden units, which encodes temporal correlations between state and control histories over a 12-step prediction horizon.

The GRU does not introduce additional physical states (e.g., suspension deflection), but instead learns a causal temporal mapping from commanded and feedback inputs to applied inputs, resulting in smooth, rate-limited actuation behavior. The GRU output is flattened and passed through two fully connected layers of 184 neurons each, followed by Mish activation and Batch Normalization to promote smooth gradients and numerical stability during training. A final *guard layer* projects the latent representation into 17 bounded physical parameters

using a Sigmoid activation, ensuring that all learned coefficients remain within physically meaningful limits (e.g., tire stiffness, damping, and inertia bounds). Training is performed end-to-end using the Adam optimizer with a starting learning rate of 10^{-3} , a batch size of 128, cosine warm-up learning-rate scheduling, and a mean-squared-error (MSE) loss objective. This configuration achieves an effective balance between dynamic accuracy and computational efficiency, enabling sub-millisecond inference while preserving generalization across both simulated and real driving domains.

B. Enhanced Physics-Based Vehicle Dynamics

The physics component of LE-PAVD is governed by a set of interpretable differential equations derived from classical vehicle dynamics. These equations form the backbone of the simulation, enabling accurate prediction of a vehicle's motion under diverse operational conditions. Our LE-PAVD integrates nonlinear tire behavior, dynamic load transfer, and suspension effects, which are solved through forward Euler integration. The key modified dynamics modeled include:

1) *Tire Slip Angle Calculations*: The slip angles α_f and α_r quantify the angular difference between the orientation of the tire and its actual direction of travel, which is crucial for lateral tire force generation. They are computed as:

$$\alpha_f = \delta - \arctan\left(\frac{l_f \omega + v_y}{|v_x|}\right) + S_{hf}, \quad (5)$$

$$\alpha_r = \arctan\left(\frac{l_r \omega - v_y}{|v_x|}\right) + S_{hr}. \quad (6)$$

This formulation accounts for combined vehicle motion due to yaw and lateral dynamics, resulting in a more accurate slip estimation.

2) *Load Transfer Effects*: During acceleration and braking, the vertical load on each axle shifts due to inertia and suspension geometry. We model longitudinal load transfer in the bicycle model via static equilibrium of vertical forces and pitch moments about the CG.

The longitudinal force-induced acceleration is modeled as,

$$a_x^{(*)} \triangleq \frac{F_{rx}}{m} \quad (7)$$

where the longitudinal force F_{rx} on the rear wheels (RWD) is given as

$$F_{rx} = (C_{m1} - C_{m2}v_x)T - C_{r0} - C_{r2}v_x^2, \quad (8)$$

i.e., the engine/brake minus resistive terms, consistent with our implementation. Solving yields

$$F_{fz} = \frac{mg l_r - h_{cg} m a_x^{(*)}}{l_f + l_r}, \quad (9)$$

$$F_{rz} = \frac{mg l_f + h_{cg} m a_x^{(*)}}{l_f + l_r}. \quad (10)$$

Equations (9)–(10) increase front-axle load under braking ($a_x^{(*)} < 0$) and rear-axle load under acceleration ($a_x^{(*)} > 0$), improving realism of the tire forces. In the single-track abstraction, lateral (left–right) transfer cancels at the axle level and is not considered in this work. We also enforce non-negativity via a small clamp in code to avoid numerical issues at very low loads.

3) *Lateral Tire Forces (Pacejka Model)*: The lateral tire forces are modeled using a load-scaled Pacejka formulation with curvature:

$$F_{fy} = \left[S_{yf} + D_f \sin(C_f \psi_f) \right] \frac{F_{fz}}{F_{z0}} \quad (11)$$

$$F_{ry} = \left[S_{yr} + D_r \sin(C_r \psi_r) \right] \frac{F_{rz}}{F_{z0}} \quad (12)$$

where

$$\psi_f = \arctan[B_f \alpha_f - E_f(B_f \alpha_f - \arctan(B_f \alpha_f))], \quad (13)$$

$$\psi_r = \arctan[B_r \alpha_r - E_r(B_r \alpha_r - \arctan(B_r \alpha_r))]. \quad (14)$$

Here F_{z0} is the nominal vertical load. The scaling (F_z/F_{z0}) makes lateral forces load-dependent. Combined slip effects (e.g., longitudinal-lateral coupling) are omitted in our formulation due to the planar simplification of the single-track model. Similarly, lateral load transfer is not modeled, as the single-track abstraction aggregates left and right tires at each axle, causing lateral forces to cancel at the axle level. These simplifications enable efficient learning and inference, while still capturing the dominant nonlinearities relevant for aggressive driving.

4) *Rate-Limited Actuator Inputs*: In real systems, steering and throttle actuators exhibit finite response and cannot change instantaneously. Rather than enforcing explicit slew-rate limits as

$$\tilde{\mathbf{U}} = \text{sat}(\mathbf{U}_{t-1} + \Delta \mathbf{U}_t^{\text{cmd}}), \quad (15)$$

LE-PAVD learns this rate-limited behavior by processing both feedback and commanded control signals through its recurrent GRU layer:

$$\tilde{\mathbf{U}} = g_\theta([\mathbf{U}_{t-k:t}^{\text{cmd}}, \mathbf{U}_{t-k:t}^{\text{fb}}]), \quad (16)$$

where $g_\theta(\cdot)$ denotes the learned temporal mapping of actuator dynamics. This formulation captures finite-rate actuator effects without hand-tuned saturation or lag parameters, preserving full differentiability for end-to-end learning. The continuous-time dynamics are discretized with forward Euler at step Δt :

$$\mathbf{X}_{t+1} = \mathbf{X}_t + \Delta t \dot{\mathbf{X}}_t, \quad (17)$$

where $\tilde{\mathbf{U}}$ are the applied, rate-limited inputs from Sec. IV-B4. This discretization keeps single-step rollouts consistent with the physics and stabilizes multi-step prediction.

V. RESULTS AND EVALUATION

This section presents a comparative analysis between DDM and LE-PAVD. The evaluation focuses on parameter efficiency, predictive accuracy, and real-time feasibility.

A. Evaluation Methods

To evaluate the open-loop predictive performance of the LE-PAVD and DDM models, we conducted experiments on two distinct datasets: a simulated dataset $\mathcal{S}_\mathcal{D}$ and a real-world dataset $\mathcal{R}_\mathcal{D}$. To produce valid comparisons we trained both models on the same train subset and tested on the same unseen test subset for both simulated and real scenarios.

The real-world dataset $\mathcal{R}_\mathcal{D}$ was collected from a full-scale autonomous Indy racecar [23]. Vehicle states were recorded at 25Hz using an Extended Kalman Filter (EKF) running at 100Hz, which fused sensor data from dual RTK-GNSS receivers and onboard inertial measurement units (IMUs). The training subset, $\mathcal{R}_\mathcal{D}^{\text{train}}$, consists of 62419 samples recorded over three laps at the Putnam Park Road Course in Indiana, while the completely unseen test subset, $\mathcal{R}_\mathcal{D}^{\text{test}}$, includes 56602 samples collected over two laps from the Las Vegas Motor Speedway.

The simulated dataset $\mathcal{S}_\mathcal{D}$ was generated using a 1:43 scale autonomous racecar simulator [24]. This data was sampled at 50Hz using a pure-pursuit controller to follow a predefined raceline across two tracks. Track 1 (ETHZ) was used to generate the training data $\mathcal{S}_\mathcal{D}^{\text{train}}$, while Track 2 (ETHZMobil) was solely used for the test set $\mathcal{S}_\mathcal{D}^{\text{test}}$. Model performance was first evaluated using one-step prediction metrics. Specifically, we computed the RMSE and the maximum error (ϵ_{max}) for three key vehicle states: $[v_x, v_y, \omega]^\top$.

Beyond single-step predictions, we assessed each model's ability to forecast multi-step future states by simulating their respective dynamics using learned parameters. From these rollouts, we computed ADE, FDE, consistent with trajectory forecasting benchmarks such as those described in [25], using a 300 ms prediction horizon for simulated data and 600 ms for real-world rollouts. We also compared neural model architecture complexity in terms of the number of model-parameters, floating point operations (FLOPs-in millions), and average inference time to evaluate computational efficiency.

All rollouts use the *applied* inputs $\tilde{\mathbf{U}}$ produced by the learned GRU mapping in Sec. IV-B4; no hand-tuned rates or bounds are used. While explicit rate-limiting equations can be integrated in future work, the current architecture captures actuator smoothness through its recurrent temporal structure. All experiments were performed on a workstation equipped with an Intel i9-14000k processor (24 cores, 32 threads, 5.70 GHz), an NVIDIA GeForce RTX 4070 SUPER GPU, and 64 GB of DDR5 memory.

B. Model Comparison with Comparable Performance

Table III summarizes the simulated and real-world coefficient ranges along with the learned mean for the enhanced dynamic single-track vehicle model described in Sec. IV-B. It includes critical parameters such as the Pacejka tire coefficients (B , C , D , E), slip angle and force shift terms (S_h , S_v), motor and resistance parameters (C_{m1} , C_{m2} , C_{r0} , C_{r2}), and the yaw moment of inertia (I_z). These bounds not only ensure physical consistency during parameter learning but also reflect the variability encountered in high-fidelity real-world data, such as those from Formula-class vehicles. By incorporating these constraints, our model maintains realism while achieving strong

predictive performance, making it suitable for both simulation-based validation and deployment in real driving scenarios.

While open-loop prediction accuracy is evaluated to isolate the effect of enhanced dynamics, the proposed architecture is explicitly designed to be control-ready. Its compact parameterization and low inference latency facilitate seamless integration into closed-loop control frameworks. As demonstrated in Section V-D, these modeling improvements translate into measurable closed-loop benefits, including shorter lap times when LE-PAVD is used as the internal dynamics model within an NMPC controller.

C. Open loop experiments

1) *Generalization Testing on Simulated Race Track:* In simulation, the ETHZ track is used exclusively for training, while the ETHZMobil track is a distinct and unseen test track used only to evaluate generalization performance. To evaluate the open-loop predictive performance of our LE-PAVD model, we benchmark it against the baseline DDM [9] using simulated data as summarized in Table IV. Despite using only 0.917M parameters, 68% fewer than DDM, LE-PAVD achieves comparable or improved performance across key metrics.

LE-PAVD yields RMSE values that are either on par with or better than those of the DDM, indicating accurate and stable predictions over the open-loop rollout. Additionally, LE-PAVD maintains low peak deviations outperforming DDM in all categories. Displacement-based metrics such as ADE and FDE further underscore the effectiveness of LE-PAVD on the simulated dataset, indicating strong physical consistency and accurate long-horizon forecasting in a fully observable simulation environment. We also assess architectural efficiency through FLOPs and inference time. LE-PAVD achieves a 3 $\times$ reduction in FLOPs (1.82M vs. 5.74M) and faster inference (0.0007s vs. 0.0010s), demonstrating superior computational efficiency without compromising predictive fidelity.

2) *Performance Evaluation on Real Race Track:* We evaluated the generalization capability of our enhanced model (LE-PAVD) using real-world vehicle data collected from the Las Vegas Motor Speedway (LVMS), comparing it against the baseline DDM. As shown in Table IV, LE-PAVD consistently

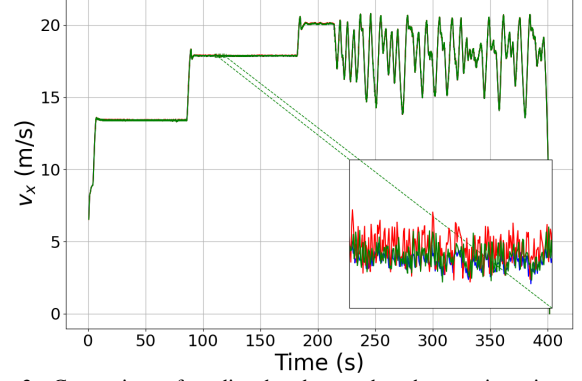


Fig. 3. Comparison of predicted and ground truth v_x trajectories on LVMS data. Ground truth is shown in blue, DDM shown in red, and LE-PAVD model in green.

improves prediction accuracy across all state variables while significantly reducing model complexity.

In terms of trajectory accuracy, LE-PAVD attains an average displacement error (ADE) of 0.0817 and a final displacement error (FDE) of 0.1442, corresponding to reductions of 16.1% and 20.6% relative to DDM. Despite these gains, LE-PAVD remains lightweight, using only 69,021 parameters (vs. 85,193 for DDM), 21.6% fewer FLOPs, and achieving faster inference time (0.0006 s vs. 0.0010 s), making it well suited for real-time deployment.

3) *Ablation Study:* Table V presents an ablation study on the real-world dataset $\mathcal{R}_D$ to assess the contribution of individual components. The variants include the baseline DDM, an independent network architecture (INA), which uses the DDM physics with the LE-PAVD network architecture. This allows us to separate the effect of the network architecture from the effect of the enhanced physics model, a third variant that augments INA with dynamic axle load transfer, and the full LE-PAVD. Among all configurations, LE-PAVD achieves the best overall performance, underscoring the effectiveness of incorporating physics-consistent dynamics with data-driven learning improves predictive performance.

Figures 3-5 show the predicted trajectories for v_x , v_y , and ω over the LVMS real-world dataset. The LE-PAVD model demonstrates strong alignment with the ground-truth signals

TABLE III
COEFFICIENT RANGES AND LEARNED PARAMETER MEANS FOR THE DYNAMIC SINGLE-TRACK MODEL ON SIMULATED AND REAL TRACK DATA.

Simulated data														
	Statistic	B_f	C_f	D_f	E_f	B_r	C_r	D_r	E_r	I_z	S_{hf}	S_{vf}	S_{hr}	S_{vr}
Range	Min	5	0.5	0.1	-2	5	0.5	0.1	-2	1.4×10^{-5}	-0.02	-0.003	-0.02	-0.003
	Max	30	2	0.9	0	30	2	0.9	0	5.6×10^{-5}	0.02	0.003	0.02	0.003
DDM	Mean	5.71	1.17	0.192	-0.158	6.68	0.940	0.188	-0.399	2.78×10^{-5}	-0.0015	3.0×10^{-4}	-0.0040	8.4×10^{-4}
LE-PAVD	Mean	8.35	0.508	0.175	-0.471	7.74	0.569	0.119	-0.422	2.79×10^{-5}	-0.0032	0.0016	-0.0025	-3.9×10^{-4}
Real data														
	Statistic	B_f	C_f	D_f	E_f	B_r	C_r	D_r	E_r	I_z	S_{hf}	S_{vf}	S_{hr}	S_{vr}
Range	Min	5	0.5	100	-2	5	0.5	100	-2	500	-0.02	-300	-0.02	-300
	Max	30	2	1×10^4	0	30	2	1×10^4	0	2000	0.02	300	0.02	300
DDM	Mean	17.91	1.34	2985	-0.94	9.33	0.80	6990	-1.16	1994	0.0070	164.9	0.0120	-27.6
LE-PAVD	Mean	6.41	0.73	5416	-0.04	7.08	0.62	6001	-1.77	1999.7	0.0028	159.2	0.0049	163.3

TABLE IV
OPEN-LOOP MODEL PERFORMANCE

	Method	v_x (m s ⁻¹)		v_y (m s ⁻¹)		ω (rad s ⁻¹)		Displacement (m)		Total Parameters	FLOPs (M)	Avg. Inference Time (s)
		RMSE	$\epsilon_{\max}$	RMSE	$\epsilon_{\max}$	RMSE	$\epsilon_{\max}$	ADE	FDE			
$\mathcal{S}_D^{\text{test}}$	DDM	8.72×10^{-5}	2.23×10^{-4}	4.02×10^{-4}	1.22×10^{-3}	3.99×10^{-3}	4.50×10^{-2}	3.77×10^{-5}	1.15×10^{-4}	2.878M	5.74	0.0010
	LE-PAVD	1.19×10^{-4}	6.97×10^{-4}	5.43×10^{-5}	5.51×10^{-3}	1.67×10^{-4}	1.93×10^{-2}	1.98×10^{-4}	4.08×10^{-3}	0.917M	1.82	0.0007
$\mathcal{R}_D^{\text{test}}$	DDM	4.46×10^{-2}	9.45×10^{-1}	1.89×10^{-1}	5.16×10^{-1}	1.07×10^{-1}	2.07×10^{-1}	9.74×10^{-2}	1.82×10^{-1}	85,193	0.162	0.0010
	LE-PAVD	2.77×10^{-2}	8.99×10^{-1}	5.93×10^{-2}	1.82×10^{-1}	9.32×10^{-3}	3.49×10^{-2}	8.17×10^{-2}	1.44×10^{-1}	69,021	0.127	0.0006

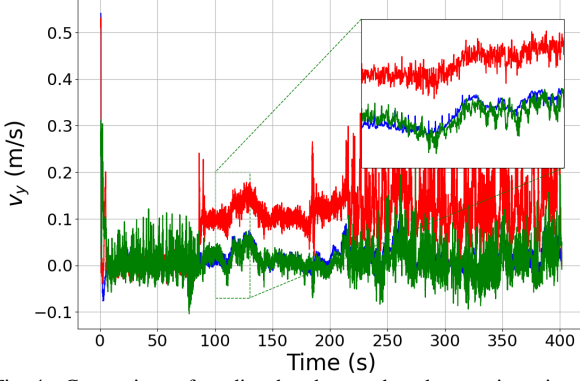


Fig. 4. Comparison of predicted and ground truth v_y trajectories on LVMS data. Ground truth is shown in blue, DDM shown in red, and LE-PAVD model in green.

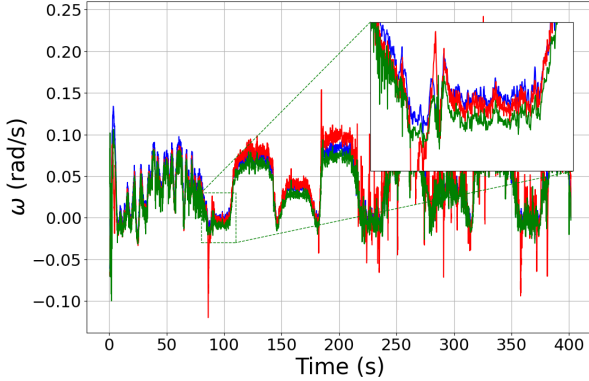


Fig. 5. Comparison of predicted and ground truth ω trajectories on LVMS data. Ground truth is shown in blue, DDM shown in red, and LE-PAVD model in green.

(blue) and outperforms DDM (red) in all three velocity components. In the zoomed-in sections, LE-PAVD (green) exhibits significantly reduced overshoot, lag, and noise, particularly during transitions in lateral motion and cornering. The model effectively captures both the amplitude and temporal trends of the dynamics with greater fidelity than DDM.

These performance improvements are achieved with significantly fewer parameters than DDM, reinforcing the benefit

of embedding physics-based priors into a lightweight neural architecture. While LE-PAVD shows strong overall performance, slightly increased sensitivity is observed during high-speed cornering and rapid transitions, likely due to the inclusion of transient dynamics.

D. Closed Loop Experiments

Table VI reports the closed-loop performance on ETHZMobil test track. All models DDM, INA and LE-PAVD, were trained on simulated data collected from ETHZ track and tested on completely unseen ETHZMobil track. In all closed-loop experiments, nonlinear model predictive controller (NMPC) is used once sufficient state history is available. During the initial warm-up phase, a pure-pursuit controller is used to generate stabilizing control inputs until the prediction horizon (15 for all models) is fully populated. This setting is identical for DDM, INA and, LE-PAVD, ensuring that all closed-loop performance differences arise solely from the underlying dynamics.

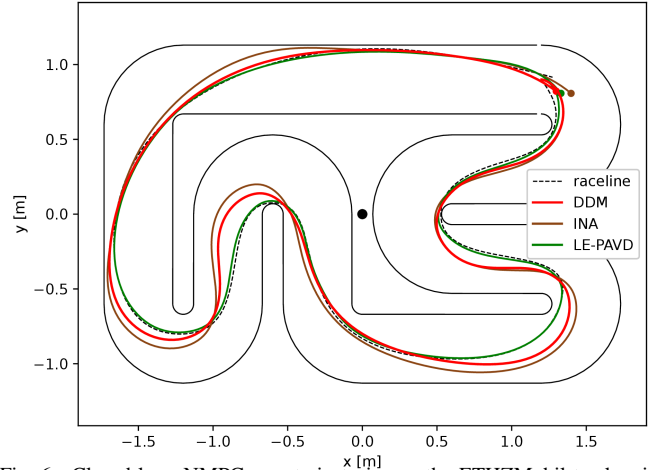


Fig. 6. Closed-loop NMPC race trajectories on the ETHZMobil track using model parameters learned from the ETHZ dataset. The dashed line denotes the reference raceline, DDM (red), INA is brown, and the proposed LE-PAVD (green).

Across all the runs, LE-PAVD completes the lap faster than the baseline DDM and INA while maintaining *zero* track violations. The ETHZMobil trajectory comparison in Fig. 6 shows that all controllers complete the lap while respecting

TABLE V
ABLATION STUDY

Method	v_x (m s ⁻¹)		v_y (m s ⁻¹)		ω (rad s ⁻¹)		Displacement (m)	
	RMSE	$\epsilon_{\max}$	RMSE	$\epsilon_{\max}$	RMSE	$\epsilon_{\max}$	ADE	FDE
Deep Dynamics Model (DDM)	0.0445	0.9464	0.1882	0.5160	0.1071	0.2065	0.0974	0.1816
Independent Network Architecture (INA)	0.0453	1.0124	0.3263	0.2651	0.1095	0.1866	0.0964	0.1974
INA + Load Transfer Effects	0.0301	0.9780	0.1502	0.4819	0.1011	0.2191	0.0804	0.1374
LE-PAVD	0.0276	0.8287	0.0592	0.1819	0.0093	0.0348	0.0817	0.1442

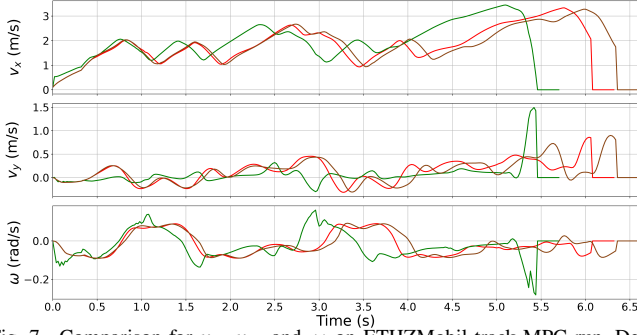


Fig. 7. Comparison for v_x , v_y , and ω on ETHZMobil track MPC run. Deep Dynamics Model (DDM) in red, INA in brown and LE-PAVD model in green.

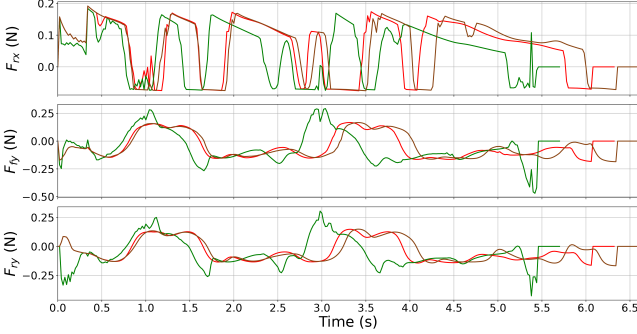


Fig. 8. Internally estimated tire force components during closed-loop NMPC run on the ETHZMobil track for DDM (red), INA in brown and LE-PAVD (green). Tire forces are model-internal quantities used by the controller, as no direct force measurements are available. LE-PAVD exhibits smoother and more structured force evolution under aggressive maneuvers.

track boundaries. However, LE-PAVD exhibits closer adherence to the nominal raceline in high-curvature segments, whereas DDM displays larger lateral deviations, particularly during aggressive cornering. The results further demonstrate that modifying the neural architecture alone as in INA, without incorporating the proposed physics-aware dynamics components. This behavior is consistent with the closed-loop state evolution shown in Fig. 7. The sharp drop to zero observed in the LE-PAVD signals near the end of the run is attributable to lap termination and logging padding rather than a loss of stability. Figure 8 presents the estimated tire-force components during closed-loop MPC operation on the ETHZMobil track, using model parameters learned from the ETHZ training dataset.

Across the lap, LE-PAVD produces smoother and more structured force profiles, with well-modulated longitudinal forces F_{rx} during acceleration and braking and bounded lateral force responses (F_{fy} , F_{ry}) through cornering maneuvers. In contrast, DDM exhibits longer force plateaus and delayed transitions in F_{rx} , along with less coordinated lateral force evolution in high-demand sections of the track. These observations indicate that the physics-aware structure of LE-PAVD enables more consistent force estimation under distribution shift, contributing to improved closed-loop tracking performance and lap-time efficiency while remaining feasible under NMPC constraints.

VI. CONCLUSION

We presented LE-PAVD, a physics-aware neural dynamics model that integrates structured vehicle dynamics with

TABLE VI
CLOSED-LOOP SIMULATION RESULTS

Track	Model	Lap Time (s)	v_x (m s^{-1})	v_y (m s^{-1})	ω (rad s^{-1})	Violations
ETHZMobil	DDM	6.3	1.78	0.12	-0.02	None
	INA	6.6	1.81	0.13	-0.02	None
	LE-PAVD	5.7	1.91	0.06	-0.03	None

learning to improve prediction and control performance near the handling limits of autonomous racing. By incorporating load-sensitive Pacejka tire forces, longitudinal load transfer, suspension-aware tire-load effects, and rate-limited actuator inputs, LE-PAVD enforces physical consistency while remaining computationally efficient. Across both simulation and real-world data, LE-PAVD matches or improves open-loop prediction accuracy relative to a deep dynamics baseline while substantially reducing model complexity and inference cost. When deployed as the internal dynamics model within an identical closed-loop framework, LE-PAVD achieves faster lap times and improved raceline adherence on unseen tracks, while maintaining zero constraint violations. These results demonstrate that enriching physical structure within a compact learned model yields measurable closed-loop performance gains beyond architectural changes alone.

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